Deriving The Schwarzschild Solution

Test Derivation

The Schwarzschild solution focuses on finding the metric for spacetime exterior to a spherically symmetric static body of mass M and radius R.

Flat metric in spherical coordinates:

$$ds^2 = \sum_{\mu\nu} g_{\mu\nu}\, dx^\mu dx^\nu = dt^2 - dr^2 - r^2 d\theta^2 - r^2 \sin^2\theta\, d\phi^2$$

We can write a metric in these coordinates in a more general form:

$$ds^2 = \sum_{\mu\nu} dx^\mu dx^\nu = U(r)\, dt^2 - V(r)\, dr^2 - W r^2 d\theta^2 - X r^2 \sin^2\theta\, d\phi^2$$

We can use our problem properties to determine U, V, W, X to an extent.

First off, our sol^n is spherically symmetric: $W = X = 1$, no θ / ϕ dep.

Static: $U = U(r)$, $V = V(r)$

Vacuum sol^n: Outside of the source of curvature, there is no matter or energy.

Thus, we have narrowed our metric down to:

$$ds^2 = \sum_{\mu\nu} g_{\mu\nu}\, dx^\mu dx^\nu = U(r)\, dt^2 - V(r)\, dr^2 - r^2 d\theta^2 - r^2 \sin^2\theta\, d\phi$$

The components of the metric and the inverse metric will be:

$$g_{00} = U,\ g_{11} = -V,\ g_{22} = -r^2,\ g_{33} = -r^2 \sin^2\theta \ ;\ g^{00} = 1/U,\ g^{11} = -1/V,\ g^{22} = -1/r^2,\ g^{33} = -1/r^2 \sin^2\theta$$

With this compact metric, we are ready to tackle the sol^n of Einstein's eq^n:

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Remember how the Ricci is just a compacted Riemann tensor:

$$R_{\mu\nu} \equiv R^\alpha{}_{\mu\nu\alpha} \ ;\ R \equiv R^\mu{}_\mu = g^{\mu\alpha} R_{\mu\alpha}.$$

Remember also that the Riemann tensor can be given in terms of Christoffel symbols:

$$R^\beta{}_{\nu\rho\sigma} = \Gamma^\beta{}_{\nu\sigma,\rho} - \Gamma^\beta{}_{\nu\rho,\sigma} + \Gamma^\alpha{}_{\nu\sigma}\Gamma^\beta{}_{\alpha\rho} - \Gamma^\alpha{}_{\nu\rho}\Gamma^\beta{}_{\alpha\sigma}$$

And the Christoffel symbols are given in terms of the metric:

$$\Gamma^\mu{}_{\nu\sigma} = \frac{1}{2} g^{\mu\lambda} \{ g_{\lambda\nu,\sigma} + g_{\lambda\sigma,\nu} - g_{\nu\sigma,\lambda} \}$$

So to find the metric, we basically just have to get the Christoffel symbols, get the Ricci, plug into eq^n to get some values of U, V's

Get the Ricci shit, put into EE to formulate O.E.s.

The Christoffel Symbols:

$$\Gamma^0_{01} = \Gamma^0_{10} = U'/2U \; ; \; \Gamma^1_{00} = U'/2V \; ; \; \Gamma^2_{12} = \Gamma^2_{21} = 1/r \; ; \; \Gamma^3_{13} = \Gamma^3_{31} = 1/r \; ; \; \Gamma^1_{11} = V'/2V$$

$$\Gamma^1_{22} = -r/V \; ; \; \Gamma^1_{33} = -r/V \sin^2\theta \; ; \; \Gamma^2_{33} = -\cos\theta\sin\theta \; ; \; \Gamma^3_{23} = \Gamma^3_{32} = \cot\theta$$

Now, the Ricci Shit:

$$R_{00} = -\frac{U''}{2V} + \frac{U'}{4}\frac{V'}{V^2} + \frac{(U')^2}{4UV} - \frac{1}{r}\frac{U'}{V} \; ; \; R_{11} = \frac{U''}{2U} - \frac{U'^2}{4U^2} - \frac{U'}{2U}\frac{V'}{2V} - \frac{V'}{Vr} \; ;$$

$$R_{22} = \frac{rU'}{2UV} + \frac{1}{V} - \frac{rV'}{2V^2} - 1 \; ; \; R_{33} = \left(\frac{rU'}{2UV} + \frac{1}{V} - \frac{rV'}{2V^2} - 1\right)\sin^2\theta = \sin^2\theta\, R_{22}$$

We can then get the equations to solve:

$$R_{00} - \frac{1}{2}g_{00}R = \frac{1}{r}\frac{V'}{V^2} + \frac{1}{r^2}\left(1 - \frac{1}{V}\right) = 0$$

$$R_{11} - \frac{1}{2}g_{11}R = -\frac{U'}{rUV} + \frac{1}{r^2}\left(1 - \frac{1}{V}\right) = 0$$

These are enough to get U and V:

$$V = \frac{1}{1 - C/r} \; ; \; U = \left(1 - \frac{C}{r}\right)$$

We can determine C by enforcing $M \to 0$ or a weak mass should give a flat metric.

This gives us:

$$ds^2 = \left(1 - \frac{2GM}{c^2 r}\right)c^2 dt^2 - \frac{dr^2}{\left(1 - \frac{2GM}{c^2 r}\right)} - r^2 d\theta^2 - r^2 \sin^2\theta\, d\phi^2$$